

MD KAMRUZZAMAN

Python Backend Developer | FastAPI | PostgreSQL | Docker | CI/CD

Dhaka, Bangladesh (Open to Remote)

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Python backend developer with a B.Tech in Computer Science and Engineering with a Machine Learning specialization. Built and deployed two practical backend and ML systems using FastAPI, PostgreSQL, Docker, CI/CD, MLflow, and Kubernetes. Experienced in REST API development, authentication, database design, automated testing, deployment, monitoring, and ML model serving. Focused on clean architecture, reliability, testing, and production-oriented backend development.

TECHNICAL SKILLS

Languages: Python, SQL, Bash

Backend: FastAPI, REST APIs, Pydantic, SQLAlchemy, Alembic

Databases: PostgreSQL, MySQL

Authentication & Security: JWT, bcrypt, OAuth2

DevOps & Cloud: Docker, Docker Compose, Kubernetes, GitHub Actions, Git, Linux, Render, Neon

Testing & Quality: pytest, FastAPI TestClient, Ruff

ML / MLOps: scikit-learn, MLflow, NumPy, Pandas

Observability: Prometheus, Grafana, Evidently

PROJECTS

Expense Tracker API

[Code](#) | [Live Demo](#)

- Built and deployed a REST API for personal expense tracking using FastAPI with JWT authentication, bcrypt password hashing, and user-scoped data isolation enforced at the query layer.
- Designed a PostgreSQL database schema using SQLAlchemy ORM and managed database migrations with Alembic.
- Implemented pagination, category and amount filtering, Pydantic validation, structured error handling, and proper HTTP status codes.
- Achieved 91% test coverage using pytest across authentication, CRUD operations, authorization, and edge cases.
- Containerized the application with Docker and configured GitHub Actions for automated testing and Ruff linting.
- Deployed the API on Render with PostgreSQL hosted on Neon and provided a public Swagger/OpenAPI interface.

ML Expense Platform

[Code](#) | [Live Demo](#)

- Built an end-to-end ML serving platform that predicts expense categories from free-text descriptions.
- Developed a scikit-learn TF-IDF and Logistic Regression pipeline and tracked four MLflow experiments with parameter sweeps for regularization strength.
- Built a FastAPI `/predict` endpoint with Pydantic validation and lazy model loading for inference.
- Containerized the application with Docker and deployed a Kubernetes configuration with two replicas, liveness and readiness probes, and self-healing behavior.
- Added Prometheus metrics for prediction counts by category and prediction latency, with Grafana dashboards for monitoring.
- Implemented Evidently-based drift detection and verified 50% column drift using simulated production data.

EDUCATION

Lovely Professional University, Punjab, India

2025

B.Tech in Computer Science and Engineering, Machine Learning Specialization

CERTIFICATIONS

- **Meta Back-End Developer** — Coursera [\[Verify\]](#)
- **Natural Language Processing Specialization** — DeepLearning.AI, Coursera [\[Verify\]](#)
- **Python Data Structures** — University of Michigan, Coursera [\[Verify\]](#)
- **Java Programming Fundamentals** — Coursera [\[Verify\]](#)